

# Qingkai Dong

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## Education

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| <b>University of Connecticut</b><br>Ph.D. in Statistics<br>Advisors: Prof. HaiYing Wang and Prof. Jun Yan                                                                                         | Aug 2023 – Expected 2027<br>Storrs, CT, USA<br>Passed the Ph.D. Qualifying Exam |
| <b>Zhongnan University of Economics and Law</b><br>MS in Mathematical Statistics<br>Advisor: Prof. Hui Zhao<br>Thesis: Model Averaging and Variable Selection for Accelerated Failure Time Models | Sep 2020 – May 2023<br>Wuhan, China                                             |
| <b>Qingdao University</b><br>BS in Applied Statistics<br>Thesis: Automatic Marking System Based on Convolutional Neural Networks                                                                  | Sep 2016 – May 2020<br>Qingdao, China                                           |

## Research Experience

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| <b>Global Biometrics Intern</b><br>Servier Pharmaceuticals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Jun 2026 – Aug 2026 |
| <ul style="list-style-type: none"><li>Designed and implemented a modular workflow to reconstruct synthetic individual participant data from published Kaplan–Meier curves and summary statistics using target-matching loss functions.</li><li>Validated close reproduction of reported survival metrics and recovery of unreported quantities through simulation studies; applied the workflow to internal data.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                     |                     |
| <b>Research Assistant</b> (Academia–Industry Collaboration)<br>Servier Pharmaceuticals & University of Connecticut                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Mar 2024 – Dec 2025 |
| <ul style="list-style-type: none"><li>Coauthored a manuscript introducing a predictive-modeling–assisted interim analysis framework for censored time-to-event endpoints; manuscript under revision.</li><li>Developed a covariate-informed approach that augments interim data by predicting event times for censored participants, improving estimation of conditional power and interim decision-making. Implemented in PyTorch.</li><li>Proposed evaluation metrics for conditional power forecast accuracy and futility decision quality; conducted extensive simulations to characterize when performance improves or degrades under varied censoring, covariate informativeness/type and prediction range.</li></ul>                                                                                     |                     |
| <b>Research Assistant</b> (Statistical Methodology)<br>University of Connecticut                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Aug 2023 – Present  |
| <ul style="list-style-type: none"><li>Coauthored a methodology paper on rare-feature-aware subsampling for regression models; manuscript in preparation for journal submission.</li><li>Developed theory showing that estimation efficiency for rare binary covariate coefficients is driven by the limited number of rare observations, explaining numerical instability and slow convergence.</li><li>Introduced balanced subsampling methods that quantifies rarity to ensure adequate representation of rare features without pilot sampling, improving robustness and efficiency.</li><li>Implemented the methods in the subsampling R package; validated performance via theoretical results, simulation studies, and real-data applications; produced documentation and reproducible examples.</li></ul> |                     |
| <b>Research Assistant</b> (Social Determinants of Health, Frailty, and Accelerated Aging in Breast Cancer Survivors)<br>Jul 2024 – Aug 2024<br>Departments of Statistics and Human Development and Family Sciences, University of Connecticut                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |
| <ul style="list-style-type: none"><li>Helped frame and prepare a study examining social determinants of health, frailty, and accelerated aging among breast cancer survivors.</li><li>Conducted data retrieval, cleaning, and integration across All of Us EHR, survey, and physical-measurement data to support subsequent statistical modeling.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                     |
| <b>Cross-Disciplinary Collaboration</b> (Chemical Sensor Data Analytics)<br>Departments of Statistics and Chemistry, University of Connecticut                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Mar 2024 – May 2024 |
| <ul style="list-style-type: none"><li>Analyzed fluorescence response data from porphyrinoid sensors; applied clustering and statistical analysis to improve compound differentiation and sensor selection.</li><li>Produced interpretable summaries and visualizations in R to support experimental decision-making.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |

**Cross-Disciplinary Collaboration** (Large Language Model Explainability)  
University of Connecticut & collaborators

2025 – Present

- Coauthored a survey manuscript on time series explainability emphasizing LLM-enabled semantic explanations; under review at a top survey journal; released an accompanying benchmark repository on GitHub.

## Software

**R package subsampling:** A statistical computing toolkit for optimal subsampling in big-data modeling. Supports GLMs, quantile regression, softmax regression and rare event/rare feature regimes.

- Implemented optimized sampling probability computation targeting variance reduction and prediction accuracy, including practical routines for pilot sampling and two-step designs.
- Engineering focus: modular API, documentation/vignettes, reproducible examples, and scalable computation patterns.

## Teaching Experience

**Principal Instructor**

Jan 2026 – May 2026

University of Connecticut

- **STAT 3675Q: Statistical Computing.** Independently taught three 50-minute sections serving approximately 20 students; designed course slides and R-based homework assignments, coordinated student presentations, and graded midterm and final examinations.
- Delivered a comprehensive R/RStudio curriculum covering data acquisition and management, visualization, reproducible workflows, regression, nonparametric inference, GLMs, categorical data analysis, time series, and classification.

## Academic Presentations

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|-----------------------------------------------------------------------------------------------|------|
| <b>Invited Talk</b> - Joint Statistical Meetings (JSM)                                        | 2026 |
| <b>Invited Talk</b> - New England Statistics Symposium (NESS)                                 | 2026 |
| <b>Invited Talk</b> - ICSA Applied Statistics Symposium                                       | 2026 |
| <b>Poster Presentation</b> - New England Statistics Symposium (NESS)                          | 2026 |
| <b>Poster Presentation</b> - The Design and Analysis of Experiments (DAE)                     | 2026 |
| <b>Poster Presentation</b> - New England Rare Disease Statistics (NERDS) Workshop, Boston, MA | 2025 |
| <b>Poster Presentation</b> - Dahshu Data Science Symposium, Storrs, CT                        | 2025 |

## Academic Service

**Journal Reviewer:** *Computational Statistics and Data Analysis*; *Statistical Papers*; *Sankhya B*; *Journal of Systems Science and Mathematical Sciences (Chinese)*

## Awards

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| <b>UConn Graduate School Conference Participation Award</b>               | Jul 2026   |
| <b>New England Statistics Symposium (NESS) Poster Award</b>               | 2026       |
| <b>ICSA Applied Statistics Symposium Student Paper Award</b>              | 2026       |
| <b>Predoc Fellowship</b> - University of Connecticut                      | 2025       |
| <b>First-class Scholarship</b> - Zhongnan University of Economics and Law | 2020, 2022 |

## Technical Skills

**Statistical Methods:** survival analysis; interim analysis & conditional power; design simulation; big-data subsampling; model averaging; variable selection

**Programming:** R (package development), Python (machine learning / deep learning), Git, LaTeX

## Selected Papers

**Dong Q**, Yan J, Zhang Q, Wang H. Preserving Rare Features in Big Data Regression: Balanced Subsampling. *In preparation for submission.*

**Dong Q**<sup>†</sup>, Man A<sup>†</sup>, Hua Z, Teng Z, Wang H, Yan J. Predictive-Modeling-Assisted Interim Decision-Making in Adaptive Trials with Censored Survival Outcomes. *Under revision.* <sup>†</sup>Equal contribution.

**Dong Q**, Liu B, Zhao H. Weighted Least Squares Model Averaging for Accelerated Failure Time Models. *Computational Statistics and Data Analysis*, 2023.

Zhao H, **Dong Q**. Variable Selection for Additive Hazards Model with Current Status Data. *Journal of Systems Science and Mathematical Sciences*, 2022.

Zhao H, Liu B, **Dong Q**. Jackknife Model Averaging of AFT Model with Current Status Data. *Acta Mathematicae Applicatae Sinica*, 2023.

Chen Z, Lucchesi G, **Dong Q**, Zheng X, Song D, Wen Q, Cheng W, Ni J, Luo D. From Signals to Semantics: A Survey on Time Series Explainability through a Human-Cognitive Lens. Manuscript under review at *ACM Computing Surveys*, 2025.